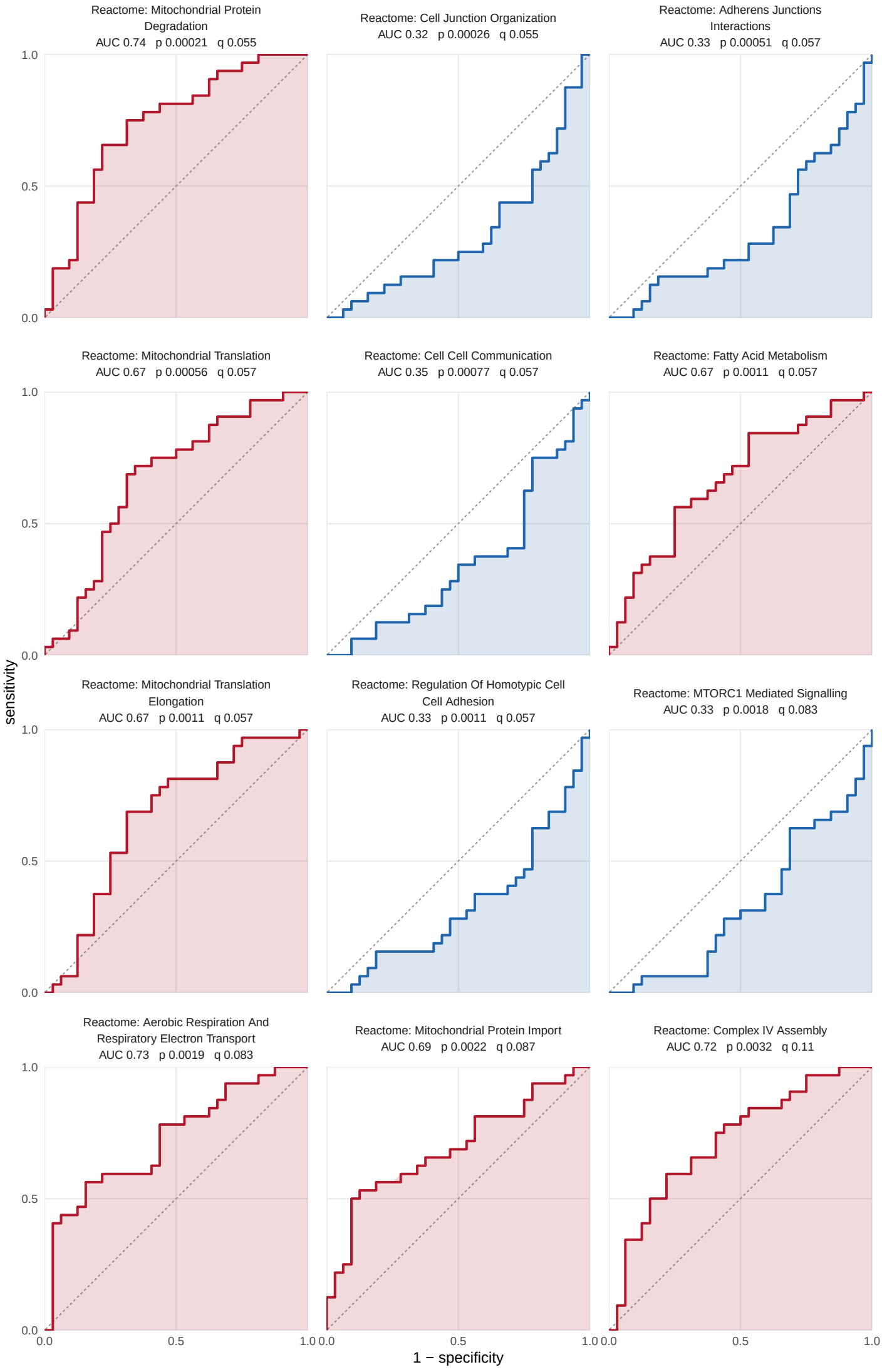
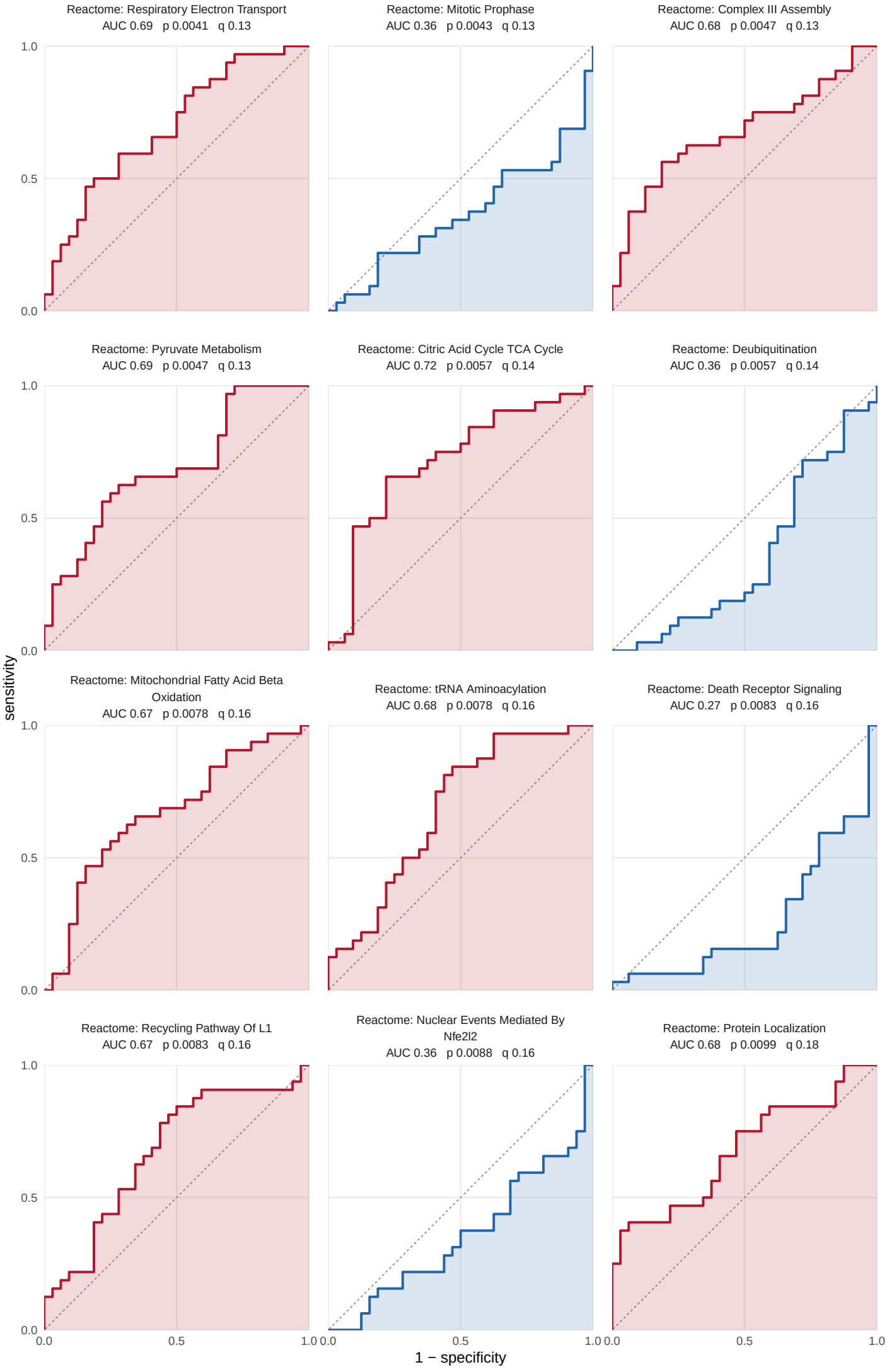
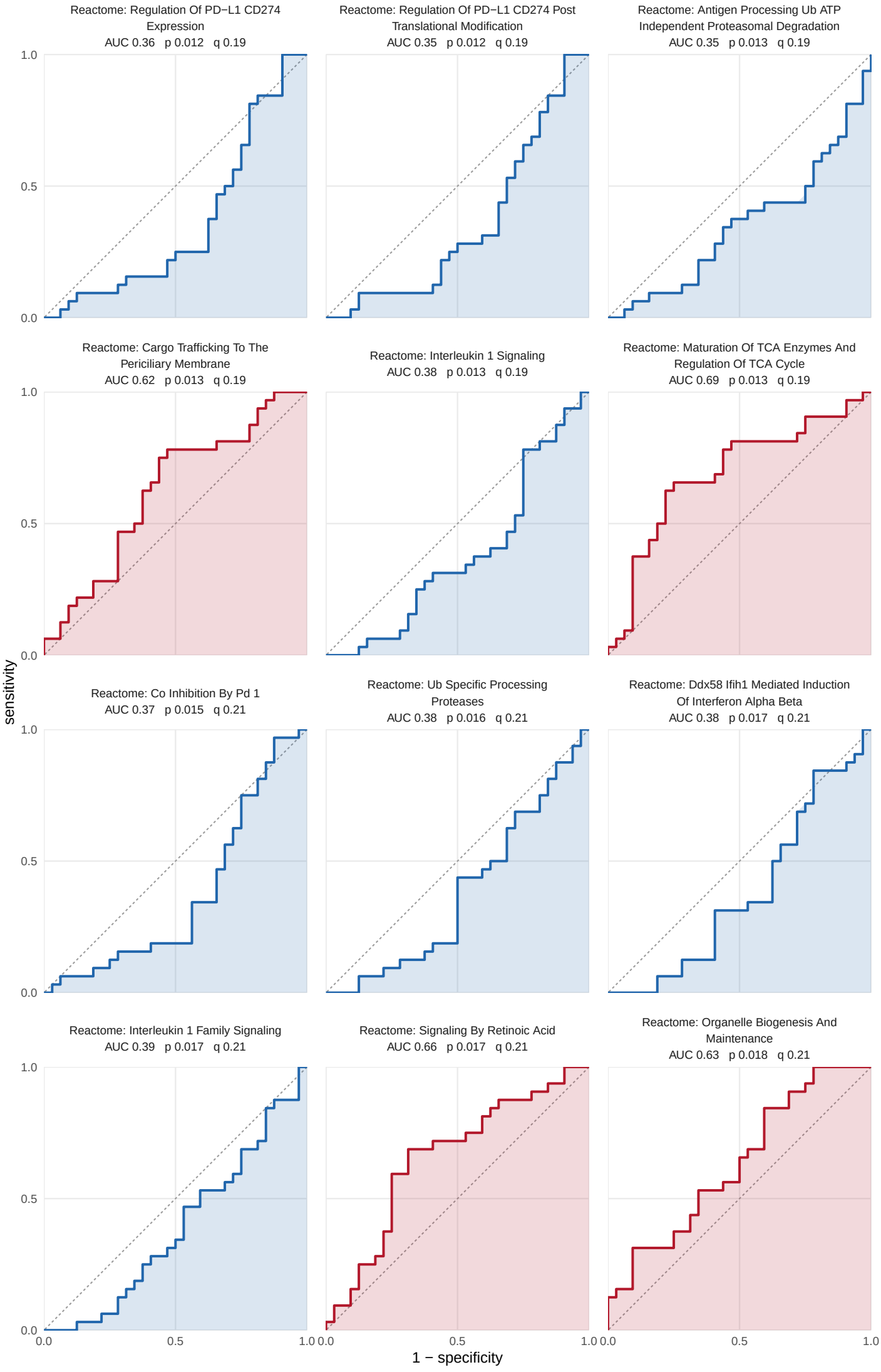
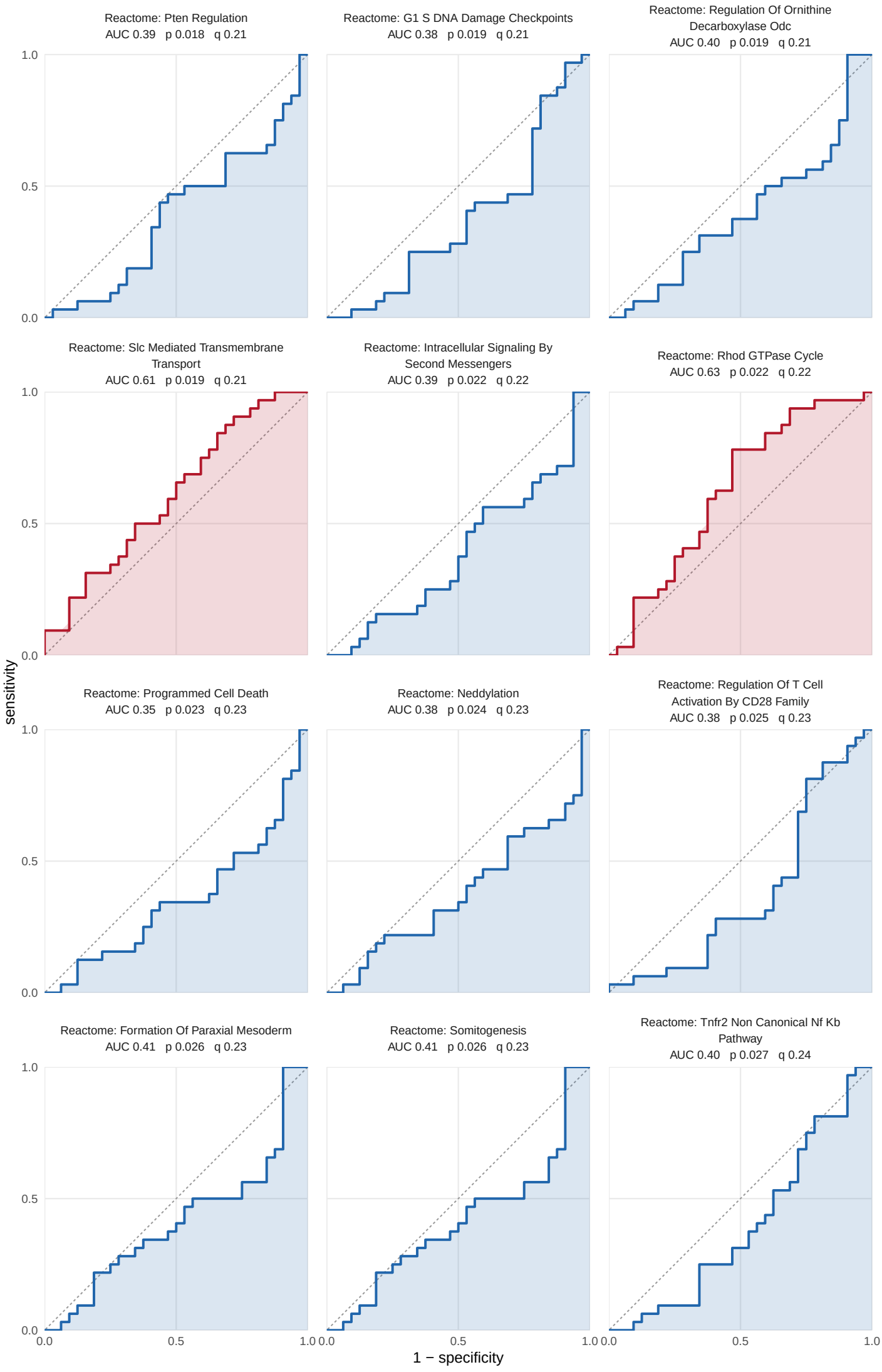
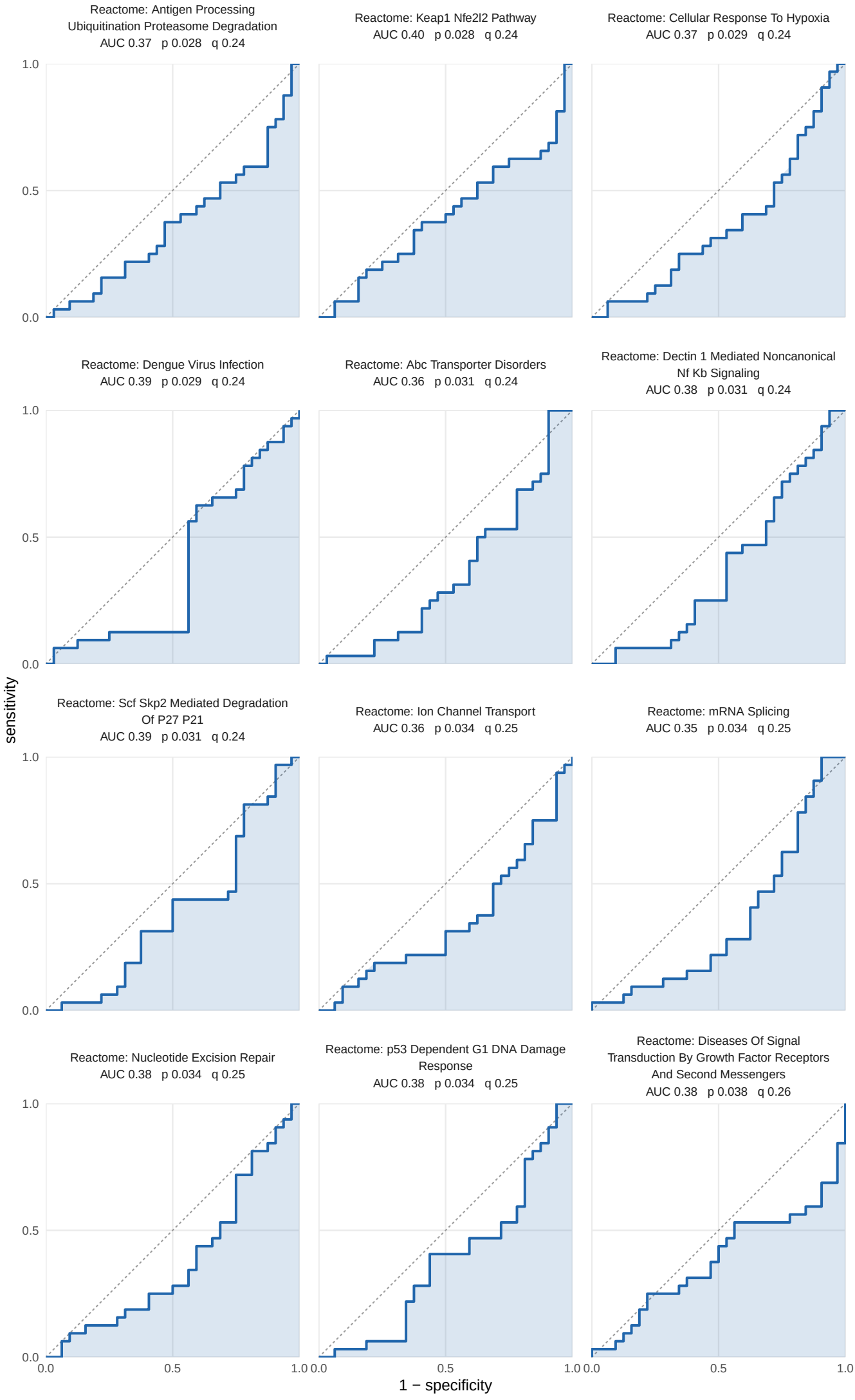
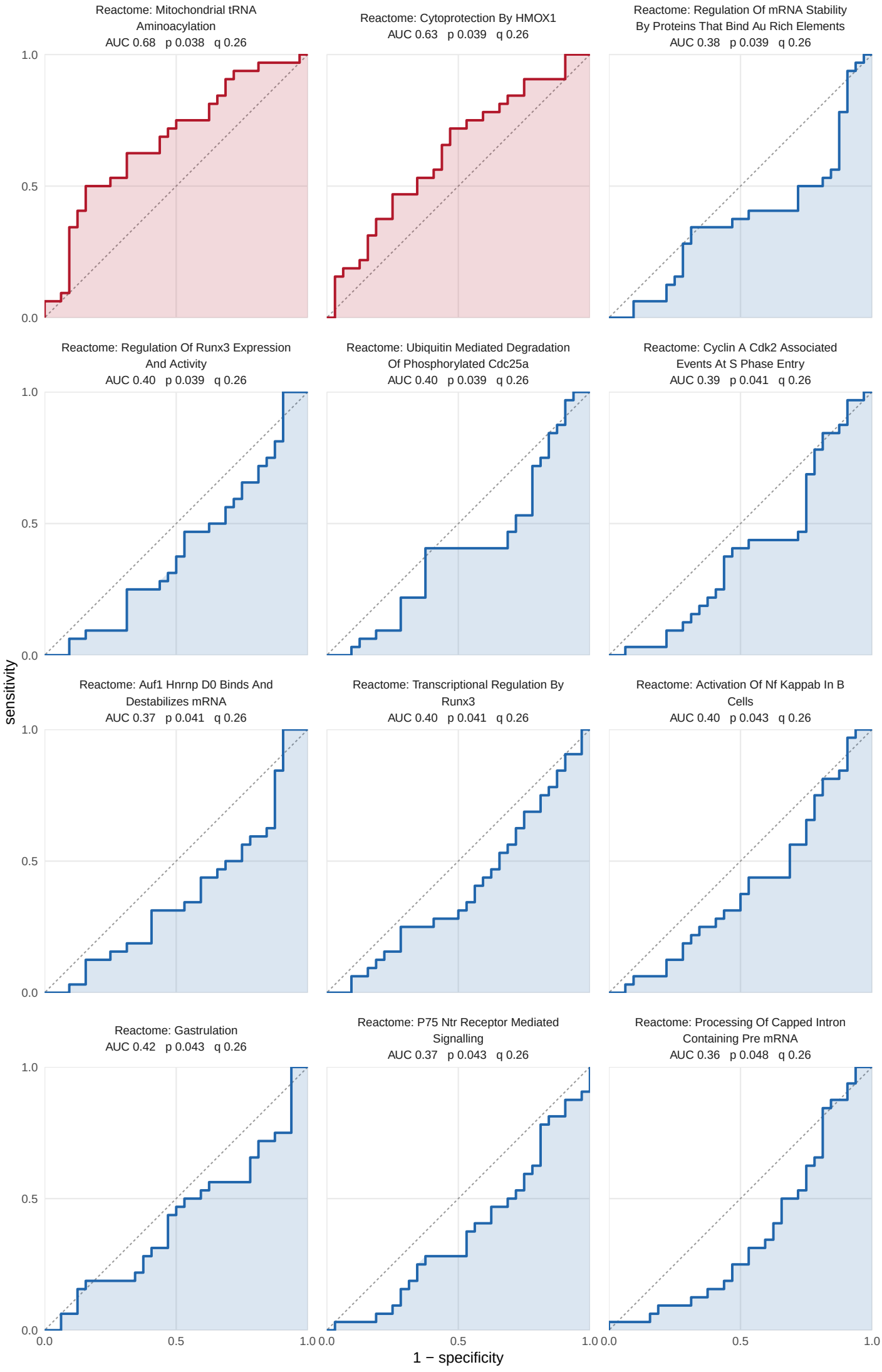

Reactome: Mitochondrial Protein Degradation
AUC 0.74 p 0.00021 q 0.055
Reactome: Cell Junction Organization
AUC 0.32 p 0.00026 q 0.055
Reactome: Adherens Junctions Interactions
AUC 0.33 p 0.00051 q 0.057
Reactome: Mitochondrial Translation
AUC 0.67 p 0.00056 q 0.057
Reactome: Cell Cell Communication
AUC 0.35 p 0.00077 q 0.057
Reactome: Fatty Acid Metabolism
AUC 0.67 p 0.0011 q 0.057
Reactome: Mitochondrial Translation Elongation
AUC 0.67 p 0.0011 q 0.057
Reactome: Regulation Of Homotypic Cell Cell Adhesion
AUC 0.33 p 0.0011 q 0.057
Reactome: MTORC1 Mediated Signalling
AUC 0.33 p 0.0018 q 0.083
Reactome: Aerobic Respiration And Respiratory Electron Transport
AUC 0.73 p 0.0019 q 0.083
Reactome: Mitochondrial Protein Import
AUC 0.69 p 0.0022 q 0.087
Reactome: Complex IV Assembly
AUC 0.72 p 0.0032 q 0.11
sensitivity
1 − specificity
0.0
0.5
1.0


Reactome: Respiratory Electron Transport
AUC 0.69 p 0.0041 q 0.13
Reactome: Mitotic Prophase
AUC 0.36 p 0.0043 q 0.13
Reactome: Complex III Assembly
AUC 0.68 p 0.0047 q 0.13
Reactome: Pyruvate Metabolism
AUC 0.69 p 0.0047 q 0.13
Reactome: Citric Acid Cycle TCA Cycle
AUC 0.72 p 0.0057 q 0.14
Reactome: Deubiquitination
AUC 0.36 p 0.0057 q 0.14
Reactome: Mitochondrial Fatty Acid Beta Oxidation
AUC 0.67 p 0.0078 q 0.16
Reactome: tRNA Aminoacylation
AUC 0.68 p 0.0078 q 0.16
Reactome: Death Receptor Signaling
AUC 0.27 p 0.0083 q 0.16
Reactome: Recycling Pathway Of L1
AUC 0.67 p 0.0083 q 0.16
Reactome: Nuclear Events Mediated By Nfe2l2
AUC 0.36 p 0.0088 q 0.16
Reactome: Protein Localization
AUC 0.68 p 0.0099 q 0.18
sensitivity
1 − specificity
0.0
0.5
1.0

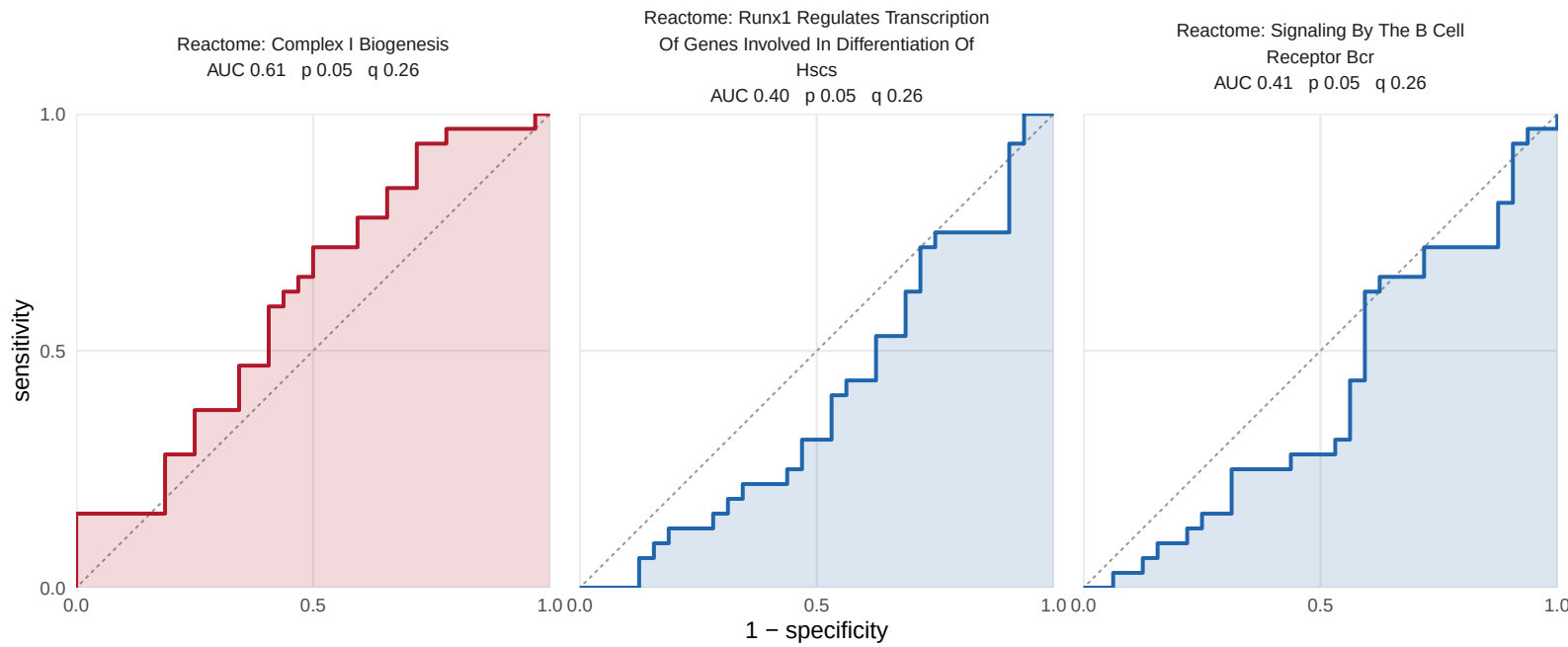

Reactome: Regulation Of PD−L1 CD274 Expression
AUC 0.36 p 0.012 q 0.19
Reactome: Regulation Of PD−L1 CD274 Post Translational Modification
AUC 0.35 p 0.012 q 0.19
Reactome: Antigen Processing Ub ATP Independent Proteasomal Degradation
AUC 0.35 p 0.013 q 0.19
Reactome: Cargo Trafficking To The Periciliary Membrane
AUC 0.62 p 0.013 q 0.19
Reactome: Interleukin 1 Signaling
AUC 0.38 p 0.013 q 0.19
Reactome: Maturation Of TCA Enzymes And Regulation Of TCA Cycle
AUC 0.69 p 0.013 q 0.19
Reactome: Co Inhibition By Pd 1
AUC 0.37 p 0.015 q 0.21
Reactome: Ub Specific Processing Proteases
AUC 0.38 p 0.016 q 0.21
Reactome: Ddx58 Ifih1 Mediated Induction Of Interferon Alpha Beta
AUC 0.38 p 0.017 q 0.21
Reactome: Interleukin 1 Family Signaling
AUC 0.39 p 0.017 q 0.21
Reactome: Signaling By Retinoic Acid
AUC 0.66 p 0.017 q 0.21
Reactome: Organelle Biogenesis And Maintenance
AUC 0.63 p 0.018 q 0.21
sensitivity
1 − specificity
0.0
0.5
1.0

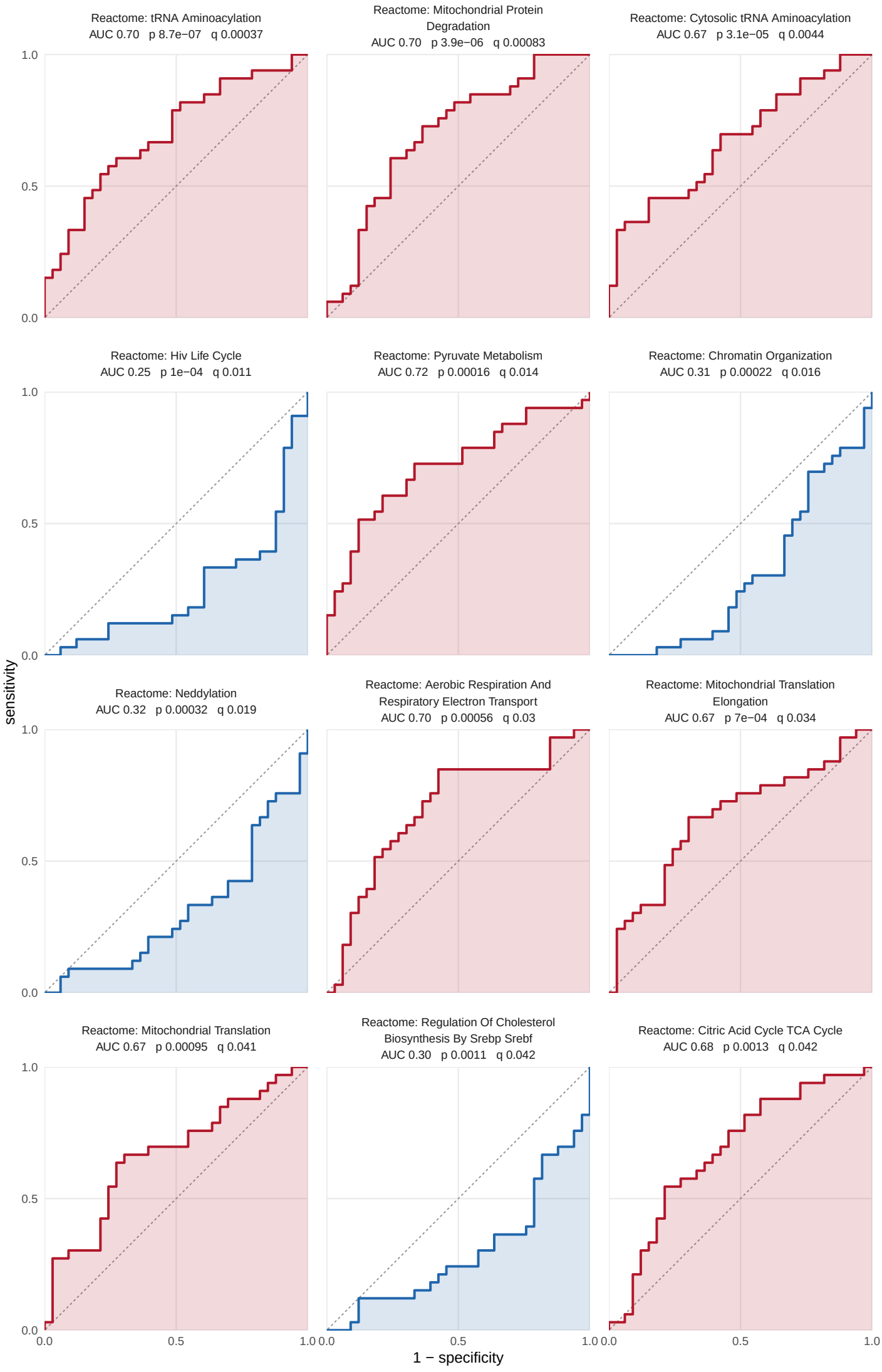

Reactome: Pten Regulation
AUC 0.39 p 0.018 q 0.21
Reactome: G1 S DNA Damage Checkpoints
AUC 0.38 p 0.019 q 0.21
Reactome: Regulation Of Ornithine Decarboxylase Odc
AUC 0.40 p 0.019 q 0.21
Reactome: Slc Mediated Transmembrane Transport
AUC 0.61 p 0.019 q 0.21
Reactome: Intracellular Signaling By Second Messengers
AUC 0.39 p 0.022 q 0.22
Reactome: Rhod GTPase Cycle
AUC 0.63 p 0.022 q 0.22
Reactome: Programmed Cell Death
AUC 0.35 p 0.023 q 0.23
Reactome: Neddylation
AUC 0.38 p 0.024 q 0.23
Reactome: Regulation Of T Cell Activation By CD28 Family
AUC 0.38 p 0.025 q 0.23
Reactome: Formation Of Paraxial Mesoderm
AUC 0.41 p 0.026 q 0.23
Reactome: Somitogenesis
AUC 0.41 p 0.026 q 0.23
Reactome: Tnfr2 Non Canonical Nf Kb Pathway
AUC 0.40 p 0.027 q 0.24
sensitivity
1 − specificity
0.0
0.5
1.0

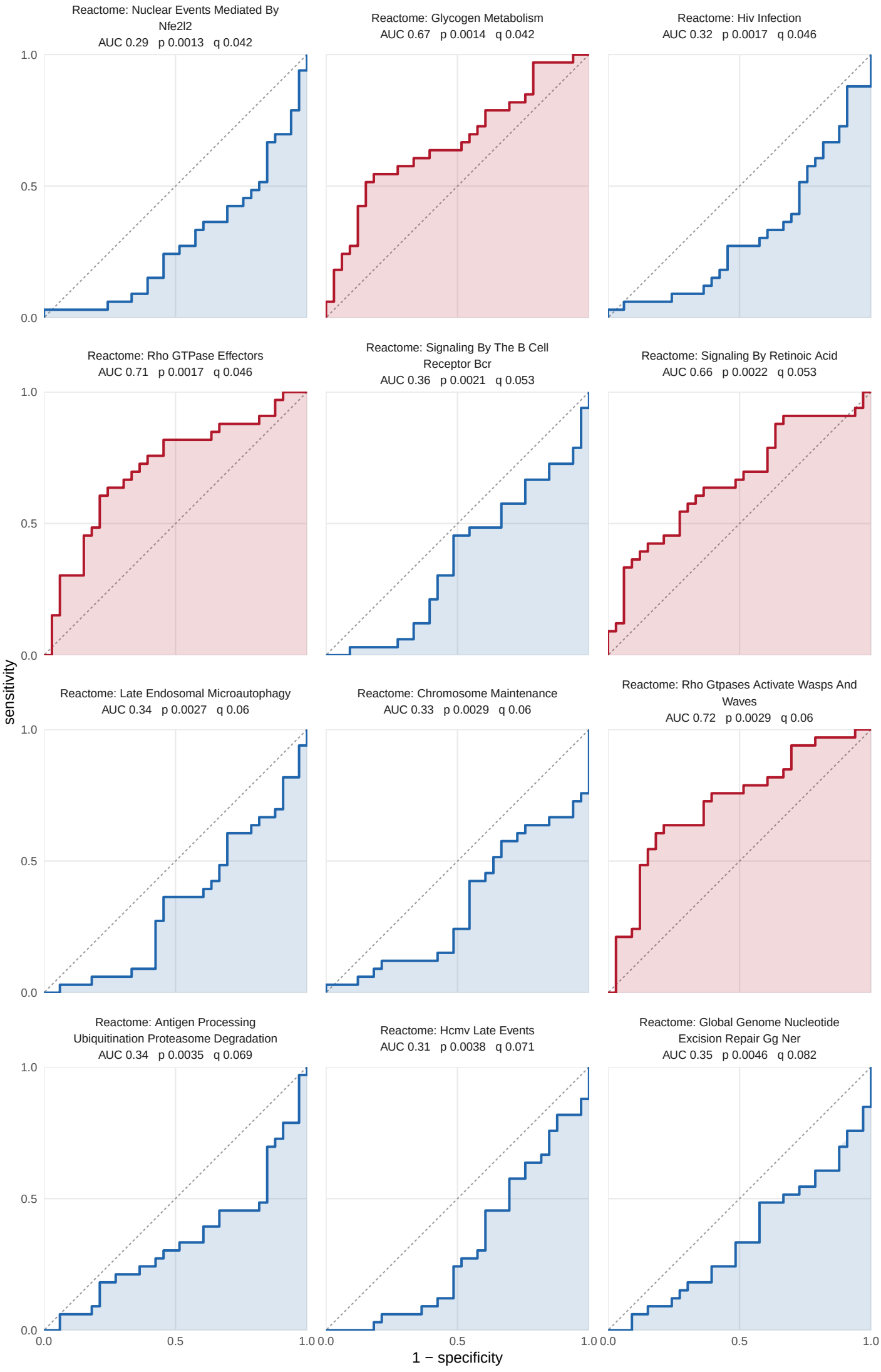

Reactome: Antigen Processing Ubiquitination Proteasome Degradation
AUC 0.37 p 0.028 q 0.24
Reactome: Keap1 Nfe2l2 Pathway
AUC 0.40 p 0.028 q 0.24
Reactome: Cellular Response To Hypoxia
AUC 0.37 p 0.029 q 0.24
Reactome: Dengue Virus Infection
AUC 0.39 p 0.029 q 0.24
Reactome: Abc Transporter Disorders
AUC 0.36 p 0.031 q 0.24
Reactome: Dectin 1 Mediated Noncanonical Nf Kb Signaling
AUC 0.38 p 0.031 q 0.24
Reactome: Scf Skp2 Mediated Degradation Of P27 P21
AUC 0.39 p 0.031 q 0.24
Reactome: Ion Channel Transport
AUC 0.36 p 0.034 q 0.25
Reactome: mRNA Splicing
AUC 0.35 p 0.034 q 0.25
Reactome: Nucleotide Excision Repair
AUC 0.38 p 0.034 q 0.25
Reactome: p53 Dependent G1 DNA Damage Response
AUC 0.38 p 0.034 q 0.25
Reactome: Diseases Of Signal Transduction By Growth Factor Receptors And Second Messengers
AUC 0.38 p 0.038 q 0.26
sensitivity
1 − specificity
0.0
0.5
1.0

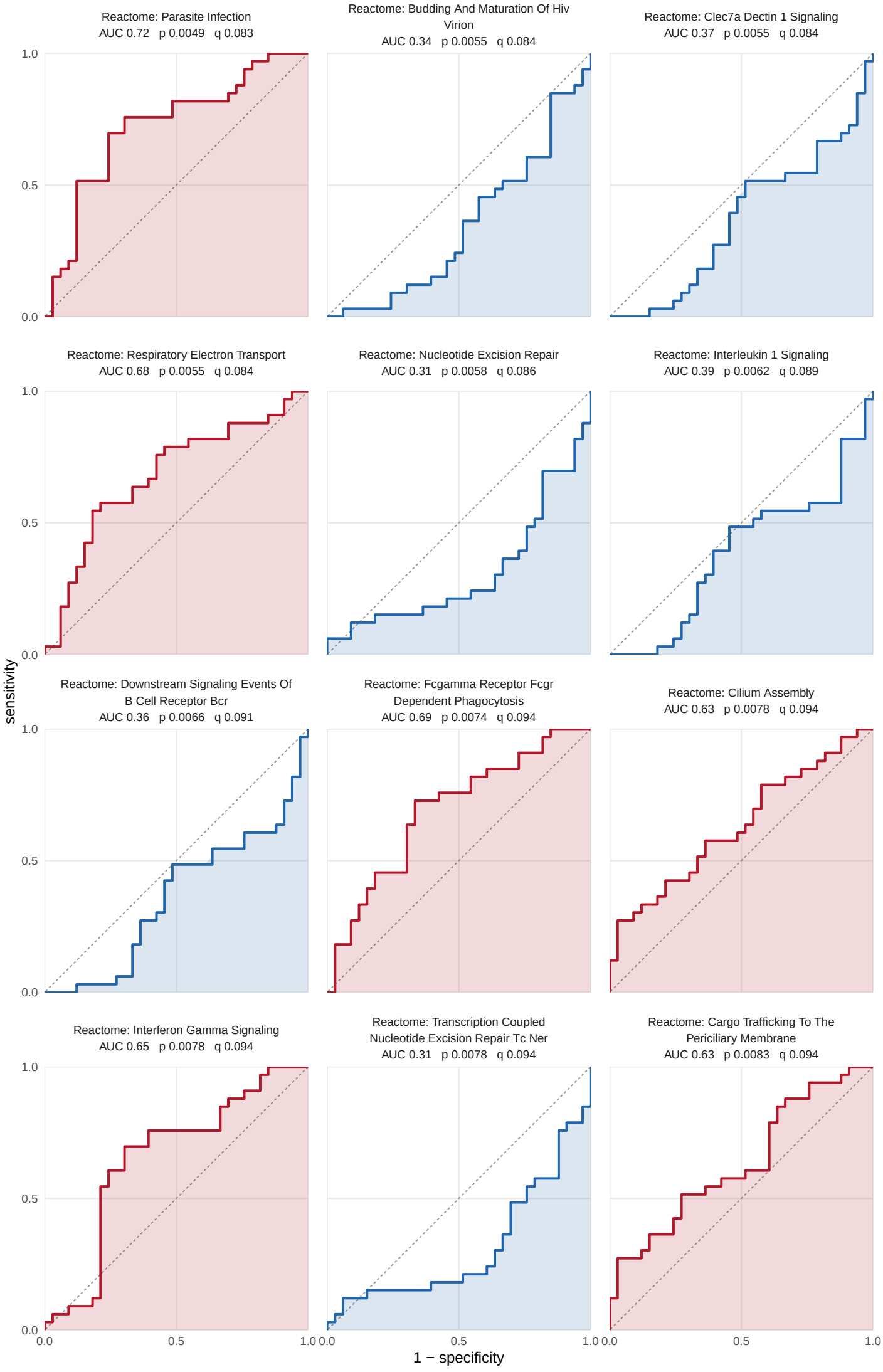

Reactome: Mitochondrial tRNA Aminoacylation
AUC 0.68 p 0.038 q 0.26
Reactome: Cytoprotection By HMOX1
AUC 0.63 p 0.039 q 0.26
Reactome: Regulation Of mRNA Stability By Proteins That Bind Au Rich Elements
AUC 0.38 p 0.039 q 0.26
Reactome: Regulation Of Runx3 Expression And Activity
AUC 0.40 p 0.039 q 0.26
Reactome: Ubiquitin Mediated Degradation Of Phosphorylated Cdc25a
AUC 0.40 p 0.039 q 0.26
Reactome: Cyclin A Cdk2 Associated Events At S Phase Entry
AUC 0.39 p 0.041 q 0.26
Reactome: Auf1 Hnrnp D0 Binds And Destabilizes mRNA
AUC 0.37 p 0.041 q 0.26
Reactome: Transcriptional Regulation By Runx3
AUC 0.40 p 0.041 q 0.26
Reactome: Activation Of Nf Kappab In B Cells
AUC 0.40 p 0.043 q 0.26
Reactome: Gastrulation
AUC 0.42 p 0.043 q 0.26
Reactome: P75 Ntr Receptor Mediated Signalling
AUC 0.37 p 0.043 q 0.26
Reactome: Processing Of Capped Intron Containing Pre mRNA
AUC 0.36 p 0.048 q 0.26
sensitivity
1 − specificity
0.0
0.5
1.0

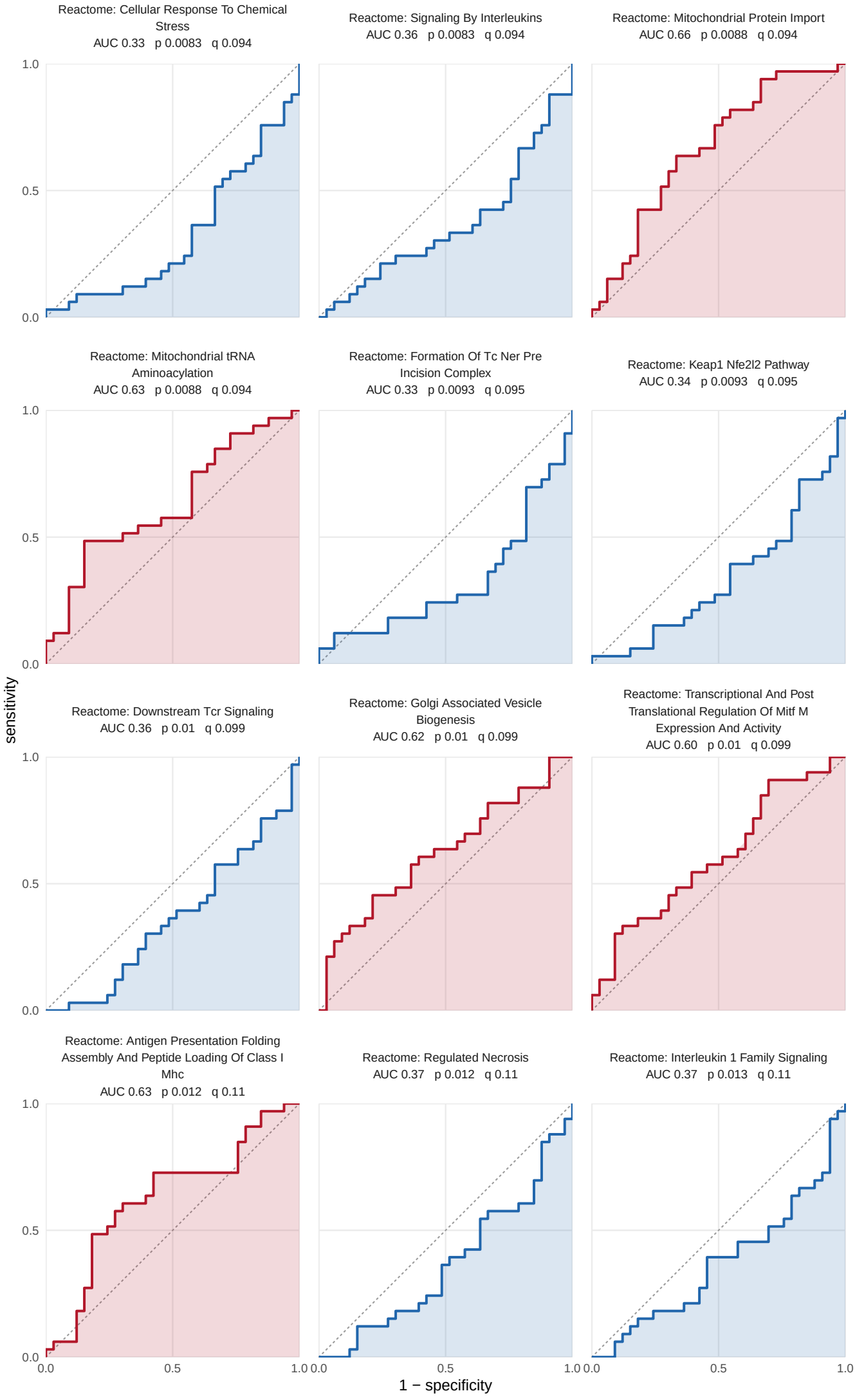
S6. ROC curves, Pre to post, BFR, Reactome (page 7 of 7)

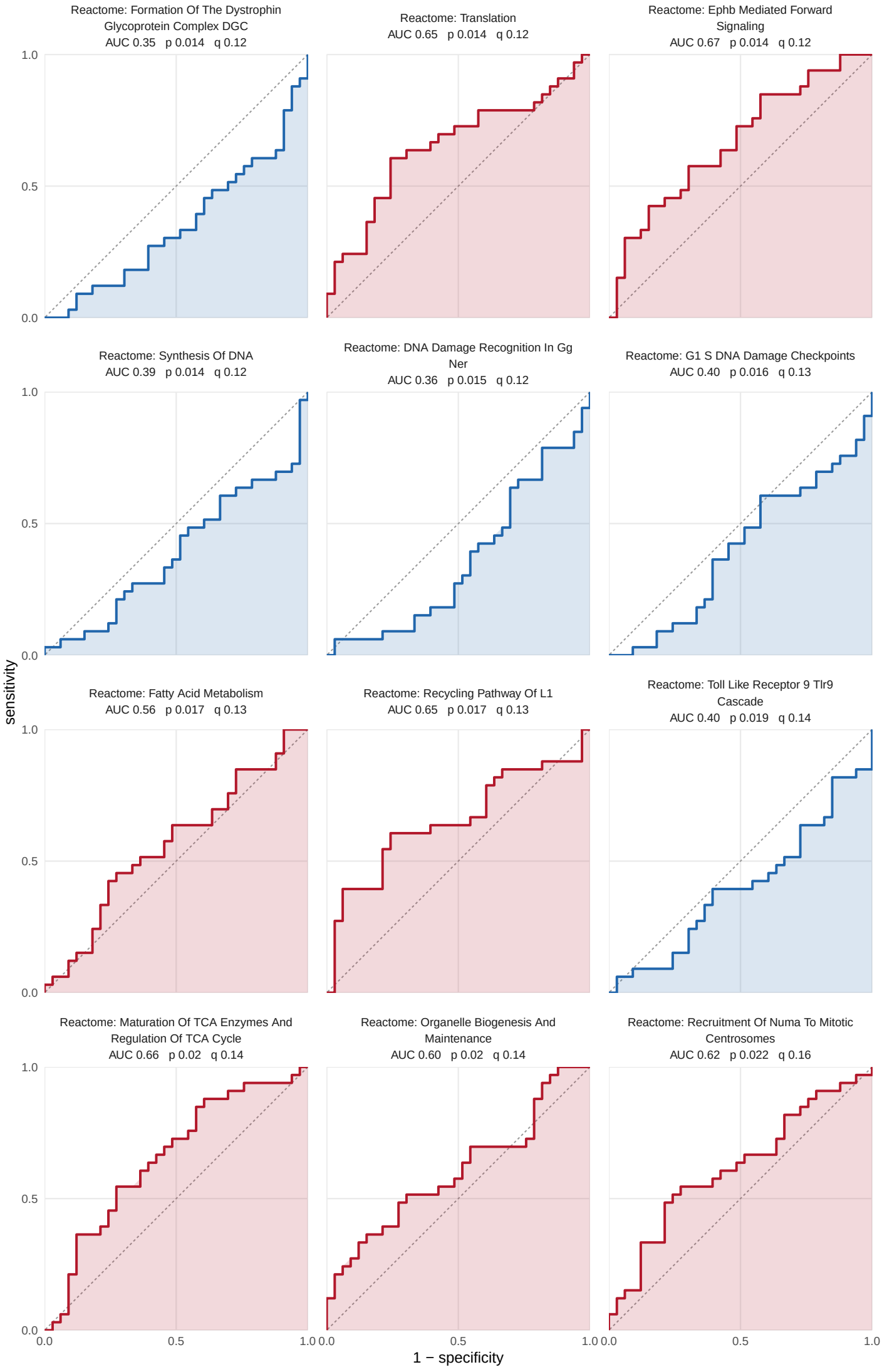


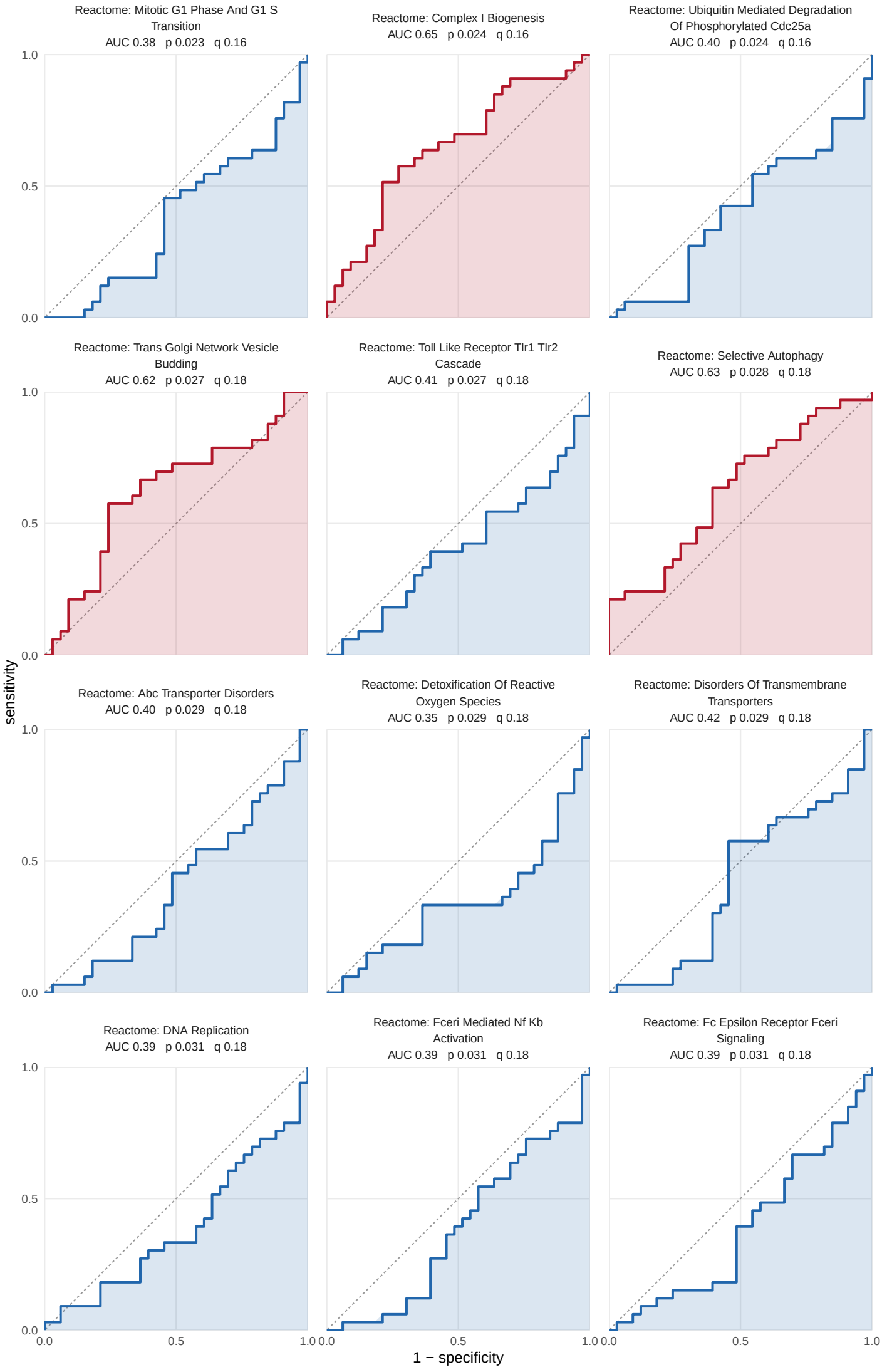

Reactome: tRNA Aminoacylation
AUC 0.70 p 8.7e−07 q 0.00037
Reactome: Mitochondrial Protein Degradation
AUC 0.70 p 3.9e−06 q 0.00083
Reactome: Cytosolic tRNA Aminoacylation
AUC 0.67 p 3.1e−05 q 0.0044
Reactome: Hiv Life Cycle
AUC 0.25 p 1e−04 q 0.011
Reactome: Pyruvate Metabolism
AUC 0.72 p 0.00016 q 0.014
Reactome: Chromatin Organization
AUC 0.31 p 0.00022 q 0.016
Reactome: Neddylation
AUC 0.32 p 0.00032 q 0.019
Reactome: Aerobic Respiration And Respiratory Electron Transport
AUC 0.70 p 0.00056 q 0.03
Reactome: Mitochondrial Translation Elongation
AUC 0.67 p 7e−04 q 0.034
Reactome: Mitochondrial Translation
AUC 0.67 p 0.00095 q 0.041
Reactome: Regulation Of Cholesterol Biosynthesis By Srebp Srebf
AUC 0.30 p 0.0011 q 0.042
Reactome: Citric Acid Cycle TCA Cycle
AUC 0.68 p 0.0013 q 0.042
sensitivity
1 − specificity
0.0
0.5
1.0

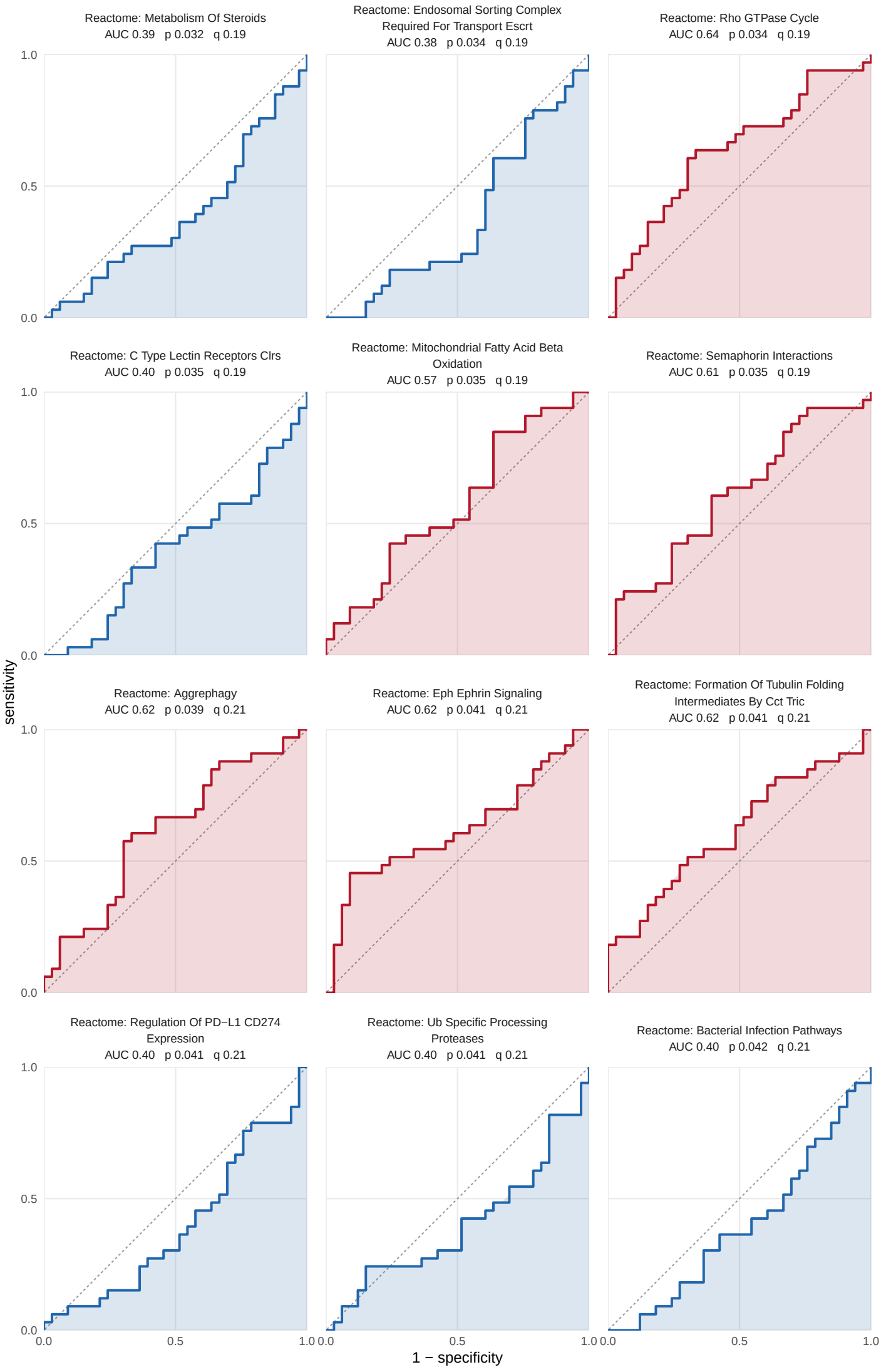

Reactome: Nuclear Events Mediated By Nfe2l2
AUC 0.29 p 0.0013 q 0.042
Reactome: Glycogen Metabolism
AUC 0.67 p 0.0014 q 0.042
Reactome: Hiv Infection
AUC 0.32 p 0.0017 q 0.046
Reactome: Rho GTPase Effectors
AUC 0.71 p 0.0017 q 0.046
Reactome: Signaling By The B Cell Receptor Bcr
AUC 0.36 p 0.0021 q 0.053
Reactome: Signaling By Retinoic Acid
AUC 0.66 p 0.0022 q 0.053
Reactome: Late Endosomal Microautophagy
AUC 0.34 p 0.0027 q 0.06
Reactome: Chromosome Maintenance
AUC 0.33 p 0.0029 q 0.06
Reactome: Rho Gtpases Activate Wasps And Waves
AUC 0.72 p 0.0029 q 0.06
Reactome: Antigen Processing Ubiquitination Proteasome Degradation
AUC 0.34 p 0.0035 q 0.069
Reactome: Hcmv Late Events
AUC 0.31 p 0.0038 q 0.071
Reactome: Global Genome Nucleotide Excision Repair Gg Ner
AUC 0.35 p 0.0046 q 0.082
sensitivity
1 − specificity
0.0
0.5
1.0


Reactome: Parasite Infection
AUC 0.72 p 0.0049 q 0.083
Reactome: Budding And Maturation Of Hiv Virion
AUC 0.34 p 0.0055 q 0.084
Reactome: Clec7a Dectin 1 Signaling
AUC 0.37 p 0.0055 q 0.084
Reactome: Respiratory Electron Transport
AUC 0.68 p 0.0055 q 0.084
Reactome: Nucleotide Excision Repair
AUC 0.31 p 0.0058 q 0.086
Reactome: Interleukin 1 Signaling
AUC 0.39 p 0.0062 q 0.089
Reactome: Downstream Signaling Events Of B Cell Receptor Bcr
AUC 0.36 p 0.0066 q 0.091
Reactome: Fcgamma Receptor Fcgr Dependent Phagocytosis
AUC 0.69 p 0.0074 q 0.094
Reactome: Cilium Assembly
AUC 0.63 p 0.0078 q 0.094
Reactome: Interferon Gamma Signaling
AUC 0.65 p 0.0078 q 0.094
Reactome: Transcription Coupled Nucleotide Excision Repair Tc Ner
AUC 0.31 p 0.0078 q 0.094
Reactome: Cargo Trafficking To The Periciliary Membrane
AUC 0.63 p 0.0083 q 0.094
sensitivity
1 − specificity
0.0
0.5
1.0

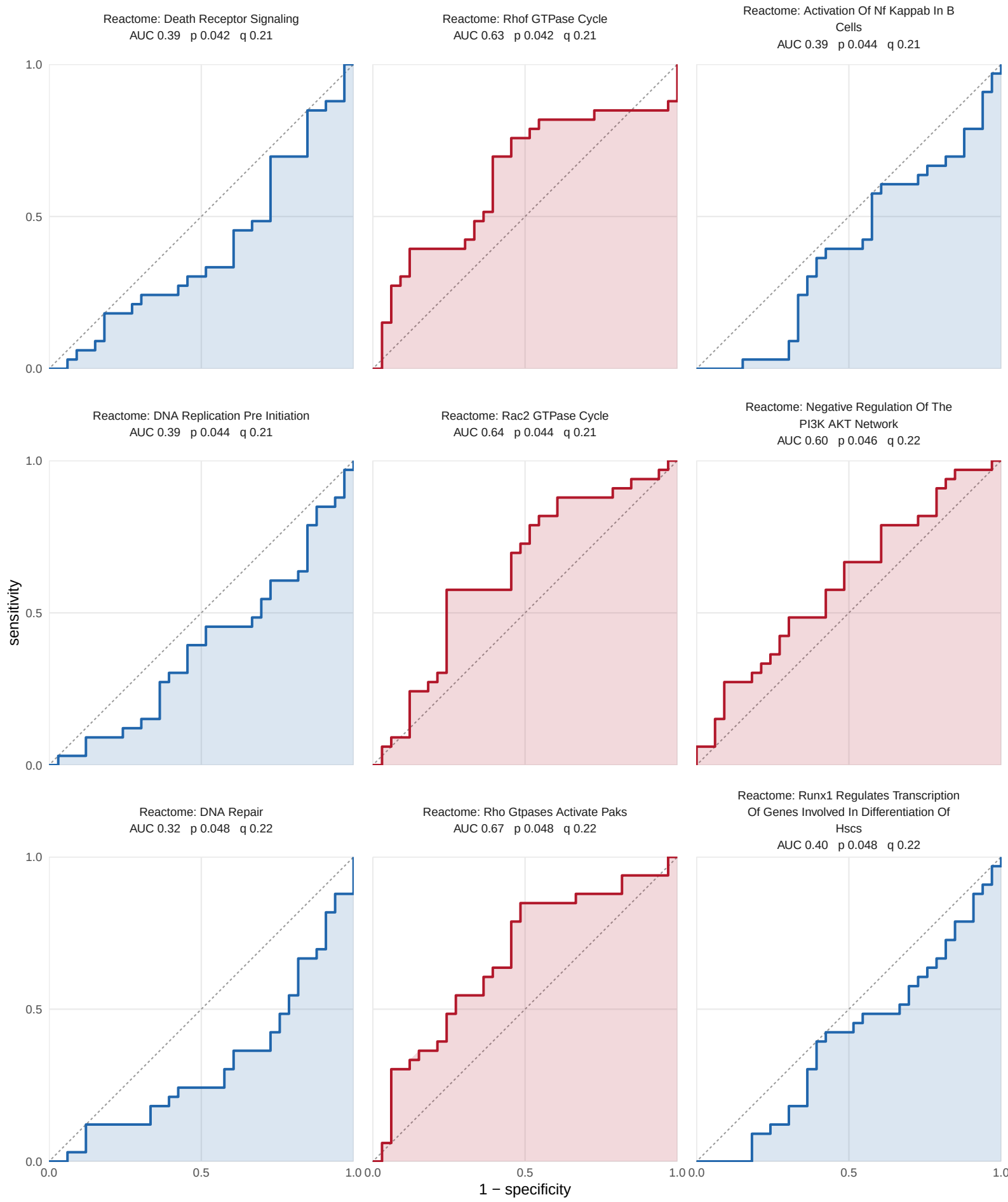

Reactome: Cellular Response To Chemical Stress
AUC 0.33 p 0.0083 q 0.094
Reactome: Signaling By Interleukins
AUC 0.36 p 0.0083 q 0.094
Reactome: Mitochondrial Protein Import
AUC 0.66 p 0.0088 q 0.094
Reactome: Mitochondrial tRNA Aminoacylation
AUC 0.63 p 0.0088 q 0.094
Reactome: Formation Of Tc Ner Pre Incision Complex
AUC 0.33 p 0.0093 q 0.095
Reactome: Keap1 Nfe2l2 Pathway
AUC 0.34 p 0.0093 q 0.095
Reactome: Downstream Tcr Signaling
AUC 0.36 p 0.01 q 0.099
Reactome: Golgi Associated Vesicle Biogenesis
AUC 0.62 p 0.01 q 0.099
Reactome: Transcriptional And Post Translational Regulation Of Mitf M Expression And Activity
AUC 0.60 p 0.01 q 0.099
Reactome: Antigen Presentation Folding Assembly And Peptide Loading Of Class I Mhc
AUC 0.63 p 0.012 q 0.11
Reactome: Regulated Necrosis
AUC 0.37 p 0.012 q 0.11
Reactome: Interleukin 1 Family Signaling
AUC 0.37 p 0.013 q 0.11
sensitivity
1 − specificity
0.0
0.5
1.0


Reactome: Formation Of The Dystrophin Glycoprotein Complex DGC
AUC 0.35 p 0.014 q 0.12
Reactome: Translation
AUC 0.65 p 0.014 q 0.12
Reactome: Ephb Mediated Forward Signaling
AUC 0.67 p 0.014 q 0.12
Reactome: Synthesis Of DNA
AUC 0.39 p 0.014 q 0.12
Reactome: DNA Damage Recognition In Gg Ner
AUC 0.36 p 0.015 q 0.12
Reactome: G1 S DNA Damage Checkpoints
AUC 0.40 p 0.016 q 0.13
Reactome: Fatty Acid Metabolism
AUC 0.56 p 0.017 q 0.13
Reactome: Recycling Pathway Of L1
AUC 0.65 p 0.017 q 0.13
Reactome: Toll Like Receptor 9 Tlr9 Cascade
AUC 0.40 p 0.019 q 0.14
Reactome: Maturation Of TCA Enzymes And Regulation Of TCA Cycle
AUC 0.66 p 0.02 q 0.14
Reactome: Organelle Biogenesis And Maintenance
AUC 0.60 p 0.02 q 0.14
Reactome: Recruitment Of Numa To Mitotic Centrosomes
AUC 0.62 p 0.022 q 0.16
sensitivity
1 − specificity
0.0
0.5
1.0

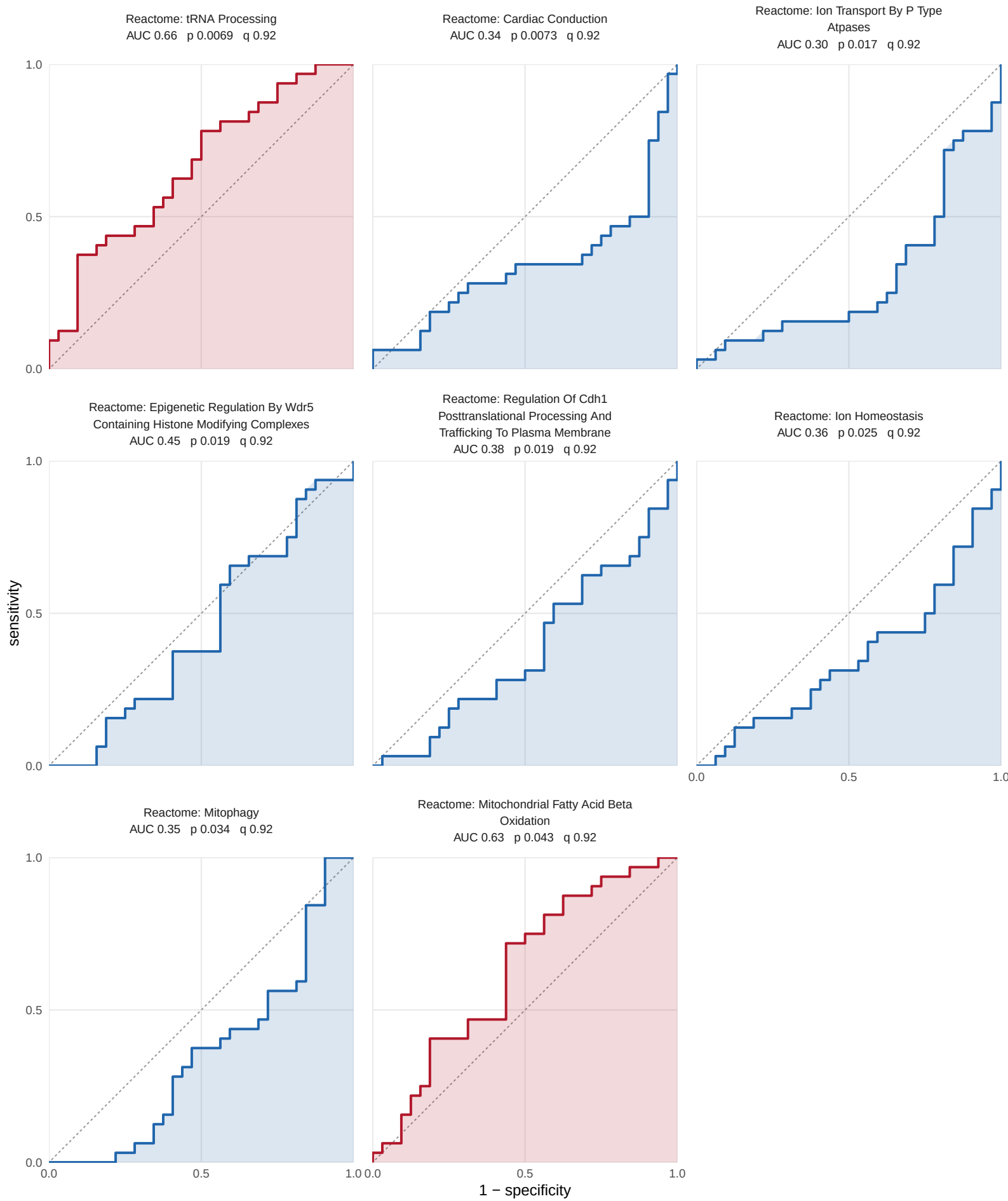

Reactome: Mitotic G1 Phase And G1 S Transition
AUC 0.38 p 0.023 q 0.16
Reactome: Complex I Biogenesis
AUC 0.65 p 0.024 q 0.16
Reactome: Ubiquitin Mediated Degradation Of Phosphorylated Cdc25a
AUC 0.40 p 0.024 q 0.16
Reactome: Trans Golgi Network Vesicle Budding
AUC 0.62 p 0.027 q 0.18
Reactome: Toll Like Receptor Tlr1 Tlr2 Cascade
AUC 0.41 p 0.027 q 0.18
Reactome: Selective Autophagy
AUC 0.63 p 0.028 q 0.18
Reactome: Abc Transporter Disorders
AUC 0.40 p 0.029 q 0.18
Reactome: Detoxification Of Reactive Oxygen Species
AUC 0.35 p 0.029 q 0.18
Reactome: Disorders Of Transmembrane Transporters
AUC 0.42 p 0.029 q 0.18
Reactome: DNA Replication
AUC 0.39 p 0.031 q 0.18
Reactome: Fceri Mediated Nf Kb Activation
AUC 0.39 p 0.031 q 0.18
Reactome: Fc Epsilon Receptor Fceri Signaling
AUC 0.39 p 0.031 q 0.18
sensitivity
1 − specificity
0.0
0.5
1.0


Reactome: Metabolism Of Steroids
AUC 0.39 p 0.032 q 0.19
Reactome: Endosomal Sorting Complex Required For Transport Escrt
AUC 0.38 p 0.034 q 0.19
Reactome: Rho GTPase Cycle
AUC 0.64 p 0.034 q 0.19
Reactome: C Type Lectin Receptors Clrs
AUC 0.40 p 0.035 q 0.19
Reactome: Mitochondrial Fatty Acid Beta Oxidation
AUC 0.57 p 0.035 q 0.19
Reactome: Semaphorin Interactions
AUC 0.61 p 0.035 q 0.19
Reactome: Aggrephagy
AUC 0.62 p 0.039 q 0.21
Reactome: Eph Ephrin Signaling
AUC 0.62 p 0.041 q 0.21
Reactome: Formation Of Tubulin Folding Intermediates By Cct Tric
AUC 0.62 p 0.041 q 0.21
Reactome: Regulation Of PD−L1 CD274 Expression
AUC 0.40 p 0.041 q 0.21
Reactome: Ub Specific Processing Proteases
AUC 0.40 p 0.041 q 0.21
Reactome: Bacterial Infection Pathways
AUC 0.40 p 0.042 q 0.21
sensitivity
1 − specificity
0.0
0.5
1.0

S7. ROC curves, Pre to post, HLRT, Reactome (page 8 of 8)



S8. ROC curves, Post BFR vs HLRT, Reactome



S9. ROC curves, Change, BFR vs HLRT, Reactome

